LESSON 4.2

Relate Fractions, Decimals, and Percentages

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can write equivalent fractions, decimals, and percents for the same value.

WHY THIS MATTERS You're designing the scoring system for a new arcade game. Players earn points as fractions of the total possible score. The leaderboard shows scores as percentages, but the game engine uses decimals. You need to convert between all three forms to build the display! A player scored 3/4 of the total possible points. Write that score as a decimal and as a percent.

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Percent *Spanish: Porcentaje*	A way to compare a number to 100, shown with the % sign.	45% = 45 out of 100
Decimal *Spanish: Decimal*	A number with a dot, like 0.5, that shows a part less than one.	0.45 = 45 hundredths
Equivalent *Spanish: Equivalente*	Having the same value, just written a different way.	3/4 = 0.75 = 75%
Benchmark *Spanish: Referencia*	A familiar number you use to guess, like 50% or 1/2.	50% = one half
Discount *Spanish: Descuento*	Money taken off the first price to make it cheaper.	$40 shirt at 25% off: discount = $10
Hundredths *Spanish: Centésimos*	The second place after the decimal point; parts out of 100.	0.09 = 9 hundredths

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

20 students voted. 7 chose green. Write the green votes as a fraction, a decimal, and a percent.

1. Fraction: 7/20.
2. 20 × 5 = 100, so 7/20 = 35/100.
3. 35/100 = 0.35 = 35%.

Answer: 7/20 = 0.35 = 35%

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Let's change 1/5 into a fraction out of 100, then a decimal and a percent.

1. _____
2. _____
3. _____
4. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- _____ / _____ is the same value as _____ written as a decimal.
- To change a decimal into a percent I _____ .
- _____ % means _____ out of every hundred.
- I know these three forms are equivalent because _____ .

Word bank percent | decimal | fraction | equivalent | hundredths | denominator | numerator | simplest form | benchmark | scale | whole | part

WATCH OUT!

A common mistake

A common mistake in Relate Fractions, Decimals, and Percents is treating the numerator and denominator as the percent's digits

LESSON 4.2

Relate Fractions, Decimals, and Percentages

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Sort these values — which are greater than 50% and which are less than 50%?

Groups: Greater than 50% | Less than 50%

_____ 3/4

_____ 0.8

_____ 65%

_____ 7/10

_____ 1/3

_____ 0.25

_____ 40%

_____ 2/5

HINT 50% is the same as 1/2 or 0.5 — that is your comparison line for every card.

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

Convert 3/5 to a percent.

(A) 35%
(B) 53%
(C) 60%
(D) 65%

HINT Percent means out of 100. What do you multiply the denominator 5 by to get 100?

Rewrite each fraction _____

Work it out _____

Simplify _____

1. **Same size pieces** *Common denominator first.*
2. **Operate** *Only then add or subtract.*
3. **Simplify** *Divide out common factors.*

3 **Circle the letter of the best answer. Show how you know.**

Convert 7/20 to a percent.

(A) 14%
(B) 20%
(C) 35%
(D) 70%

HINT The denominator 20 is friendly: 20 × 5 = 100, and percents are always 'out of 100'.

Rewrite each fraction _____

Work it out _____

Simplify _____

1. **Same size pieces** *Common denominator first.*
2. **Operate** *Only then add or subtract.*
3. **Simplify** *Divide out common factors.*

4 **Circle the letter of the best answer. Show how you know.**

Which decimal is equivalent to 45%?

(A) 0.045
(B) 0.45
(C) 4.5
(D) 45.0

HINT Percent means 'out of 100', so 45% is the fraction 45/100.



WORK

1. **Whole** *100% is what number?*
2. **Find 10%** *Divide by 10.*
3. **Build** *Add or multiply to the percent.*
4. **Check** *Is it under or over the whole?*

5 **Circle the letter of the best answer. Show how you know.**

What fraction is equal to 0.75?

(A) 3/4
(B) 3/5
(C) 7/5
(D) 7/10

HINT 0.75 is read 'seventy-five hundredths' — the place value tells you the denominator.

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 Find the mistake. Explain it, then show the correct work.

Spot the Common Mistake

1. **Convert 3/5 to a percent** 3/5
2. **Read the digits as the percent** The numerator is 3 and the denominator is 5, so the student writes 35%.
3. **Final answer** 3/5 = 35%

Which step has the mistake? Explain what went wrong, then show the correct work.

> **HINT** Do the digits 3 and 5 tell you how many parts out of 100?

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

What number did you multiply by to turn each denominator into 100, and how did that fraction out of 100 give you the percent?

SENTENCE STARTER I multiplied the top and bottom of ___ by ___ to get ___/100, so the percent is ___%.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 4.2

Relate Fractions, Decimals, and Percentages

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Complete each row. Build the fraction out of 100 first — that column hands you the percent.

Fraction	Equivalent fraction out of 100	Decimal	Percent
2/5			
		0.125	
			90%
5/8			

SHOW YOUR WORK

2 **Mark and label each value on the number line.**

Place each value on the number line: A = 3/8, B = 0.6, C = 85%



SHOW YOUR THINKING

3 **Write the name of the correct group on each line.**

Order these values from least to greatest by sorting them into the correct position.

Groups: 1st (Smallest) | 2nd | 3rd | 4th | 5th (Largest)

_____ 1/5

_____ 0.35

_____ 2/5

_____ 55%

_____ 3/4

PART 2 Apply it

real-world problems

4 **Circle the letter of the best answer. Show how you know.**

A game shows your score as 7/8 complete. What percent is that?

(A) 75%

(B) 78%

(C) 87.5%

(D) 88%

Rewrite each fraction _____

Work it out _____

Simplify _____

PART 3 Explain and correct

reasoning

5 Find the mistake. Explain it, then show the correct work.

Spot the Common Mistake

1. **Convert 5/8 to a percent** 5/8
2. **Divide the bottom by the top** 8 ÷ 5 = 1.6
3. **Multiply by 100** 1.6 × 100 = 160%
4. **Final answer** 5/8 = 160%

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

What number did you multiply by to turn each denominator into 100, and how did that fraction out of 100 give you the percent?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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Challenge · Extension

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PART 1 Understand the idea

models and meaning

1 **Mark and label each value on the number line.**

Place 1/3, 0.4, and 38% on the number line. Which is greatest?



SHOW YOUR THINKING

PART 2 Explain and correct

reasoning

2 Find the mistake. Explain it, then show the correct work.

Find the Conversion Error

1. **Convert 2/3 to a percent** 2/3
2. **Divide numerator by denominator** 2 ÷ 3 = 0.67
3. **Move decimal two places** 0.67 → 67%
4. **Final answer** 2/3 = 67%

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

3 Answer in complete sentences.

A student says 3/8 is greater than 40% because 3 and 8 are bigger numbers than 4 and 0. Is the student correct? Convert both values to the same form and explain.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

What number did you multiply by to turn each denominator into 100, and how did that fraction out of 100 give you the percent?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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